

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th May 2023

Subject SA3 – General Insurance

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** You are a Pricing Actuary and work with a multi-line non-life insurer in India. Your CEO has asked the Actuarial function to work with the Reinsurance team that is in the process of finalising the treaties for next financial year.
- i)**
- a)** As per IRDAI (Re-insurance) Regulations what are the expectations from Indian insurer on Catastrophic Risk Protection. (2)
 - b)** During annual budgeting exercise, your Reinsurance Head has mentioned that reinsurance cost is expected to increase next year. Discuss all factors that could impact reinsurance cost to increase. (6)
- ii)** Your organisation has recently appointed a new Chief Risk Officer (CRO), who is also new to the general insurance industry. A Product Management Committee (PMC) meeting has been scheduled to discuss the Standard Fire Product ‘Bharat Griha Raksha (BGR)’.
- a)** What is the role of Chief Risk Officer (CRO) in Product Management Committee. (2)
 - b)** The CRO has requested the Actuarial function to help him understand the challenges with Standard Fire Product ‘Bharat Griha Raksha (BGR)’ which covers structure as well as contents.

Draft a note for your CRO covering challenges associated with BGR product. (8)
- iii)** Your insurer has achieved less than planned growth over last few years due to capital constraints. Lower profitability over the period has made the problem worse.

Your Insurance Company is planning to raise capital to achieve growth. Briefly discuss the possible sources of capital available to insurer with special considerations as applicable. (10)
- Your Chief Financial Officer (CFO) thinks that historical catastrophic (CAT) losses observed by the insurer over last 20 years can be used to come up with Capital charge associated with Catastrophe risk.
- iv)** List the issues associated with using historical loss information to come up with Capital charge for CAT Risk. (4)
 - v)** List other data items that can help in the process to arrive at capital charge for CAT risk. (10)
- You are interviewing a Motor Underwriter who mentions that his annual bonus largely depends on Motor Own Damage (OD) Profitability seen over the Financial Year.
- vi)** Discuss the advantages and disadvantages of using Motor OD profitability as Key Performance Indicator (KPI) for Motor Underwriter including other considerations. (8)
- Q. 2)** Recently, the insurance regulator of the country in which Credence Petroleum Ltd, a leading private sector Oil and Gas company is operating, has decided to permit companies to own Captive Insurance companies.

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In view of the same, Board of the Credence Petroleum Ltd., is contemplating to start its own captive insurance company.

- i) Discuss the reasons why the company wants to start its own captive insurance company, along with the disadvantages of the suggestion. (15)
 - ii) Due to the high risk, the board of the company is proposing to have an XoL treaty with Rs 100 crores excess limit for any one loss event. Reinsurance broker collected quotes from various reinsurers for the same. The premiums quoted were found to be 65% higher than the unadjusted average losses over last 15 years of the company. However, after seeing high reinsurance premiums, the Board is examining if the excess limit needs to be increased. Discuss this idea of the Board. (10)
- [25]

Q. 3) After introduction of circular by the regulator on “Usage Based Insurance” (UBI) for motor vehicles, there are many innovative products in the market. XYZ Insurance Company Limited has introduced an UBI product, where the customer has to use the mobile app to turn on the insurance option before starting the vehicle and they can turn off the insurance option whenever the same is not required. But this is applicable only to Own Damage losses. Also, multiple cars and two wheelers owned by the same customer can be insured under a single policy.

- i) Discuss the pricing and reserving of such a product when compared to the traditional motor products. (13)
 - ii) Explain the impact of such product in ALM for Motor Line of Business of the insurer, provided insurer sells only this product in future. (2)
 - iii) Also discuss the following aspects with respect to the product: Investment return, Lapses, Expenses & Commission, Customer behaviour and Capital. (10)
- [25]
